

Detection and Response – Sound the Alarm

Comprehensive Documentation of Practical Learning Experience

Google Cybersecurity Professional Certificate

This documentation provides a comprehensive overview of the knowledge, technical skills, and practical hands-on experience acquired during the **Detection and Response – Sound the Alarm** course in the Google Cybersecurity Professional Certificate programme. The course focused on cybersecurity monitoring, threat detection, incident response, network analysis, and log investigation using real-world security tools and techniques.

The programme emphasized the importance of identifying detection signals, analyzing suspicious activities, investigating alerts, and applying appropriate response procedures to security incidents. Through multiple practical labs and simulations, I developed a deeper understanding of how cybersecurity analysts monitor systems, detect threats, and respond effectively to cyber incidents in operational environments.

1. Overview of Detection and Response

Cybersecurity detection and response are critical components of modern security operations. Organizations continuously monitor systems, endpoints, servers, and network traffic to identify suspicious activities that may indicate cyberattacks or unauthorized access attempts.

The course introduced the role of Security Operations Centers (SOCs) and how analysts use monitoring tools, security logs, intrusion detection systems, and alerting platforms to identify potential threats. I learned that rapid detection and effective response are essential for minimizing damage, protecting sensitive information, and maintaining business continuity.

2. Understanding Security Signals

One of the most important lessons from the course involved understanding cybersecurity signals and indicators of suspicious activity. Security signals are pieces of information generated by systems, networks, and applications that help analysts determine whether malicious activities may be occurring.

Examples of signals analyzed during the course included:

- Unusual network traffic patterns
- Unauthorized HTTP requests
- Unexpected login attempts
- Abnormal system behavior
- Suspicious IP communication
- Alert signatures generated by intrusion detection systems

The course demonstrated how analysts correlate these signals to identify attacks, investigate incidents, and make informed security decisions.

3. Incident Detection and Monitoring

The course emphasized continuous security monitoring as a key cybersecurity function. I learned how organizations monitor network traffic, system logs, and security alerts to detect potential attacks.

Monitoring activities included:

- Reviewing alert logs
- Investigating suspicious traffic patterns
- Analyzing intrusion detection alerts
- Monitoring network protocols
- Identifying malicious behaviors within network communications

These activities strengthened my understanding of proactive cybersecurity defense and the importance of early threat detection.

4. Hands-On Experience with Suricata

A major component of the course involved practical work with **Suricata**, an open-source intrusion detection and network monitoring system used by cybersecurity professionals to analyze network traffic and detect suspicious activities.

Through hands-on exercises, I learned how to:

- Configure Suricata for packet analysis
- Create custom detection rules
- Analyze packet capture (PCAP) files
- Generate alerts based on suspicious traffic
- Investigate logs stored in **fast.log** and **eve.json**
- Interpret JSON-formatted event data

These exercises provided practical exposure to network monitoring and real-world threat detection workflows.

5. Practical Log Analysis

The course placed significant emphasis on log analysis because logs contain valuable evidence for detecting and investigating cybersecurity incidents.

During the labs, I worked with:

- Suricata alert logs

- JSON event logs
- Packet metadata
- HTTP request information
- Source and destination IP addresses
- TCP connection details

I learned how to use Linux command-line utilities such as **cat** and **jq** to inspect, format, and analyze security logs efficiently.

6. Linux and Command-Line Operations

The course strengthened my Linux command-line skills, which are essential for cybersecurity analysis and incident response.

Commands practiced included:

```
sudo suricata -r sample.pcap -S custom.rules -k none
```

```
ls -l /var/log/suricata
```

```
cat /var/log/suricata/fast.log
```

```
cat /var/log/suricata/eve.json
```

```
jq . /var/log/suricata/eve.json | less
```

These commands were used to process packet capture files, inspect generated logs, analyze alert data, and review JSON-formatted network events. Through repeated practice, I improved my ability to investigate network activity and identify suspicious behaviors.

7. Incident Response Procedures

The course introduced the structured phases of incident response used by cybersecurity teams to manage and contain security incidents effectively.

The incident response lifecycle covered:

- Preparation

- Detection and Analysis
- Containment
- Eradication
- Recovery
- Post-Incident Review

I learned how analysts identify incidents, investigate alerts, contain threats, remove malicious activity, and restore affected systems while documenting findings for future improvements.

8. Skills and Knowledge Acquired

Through this course, I developed both technical and analytical cybersecurity skills, including:

- Threat detection and monitoring
- Security event analysis
- Log investigation and interpretation
- Intrusion detection system management
- Linux command-line proficiency
- Network traffic analysis
- Security alert investigation
- JSON data analysis
- Incident response methodologies
- Cybersecurity analytical thinking

9. Practical Lab Documentation

The following screenshots demonstrate practical lab exercises completed during the course. These exercises involved configuring custom Suricata detection rules, processing packet capture files, generating alerts, and analyzing JSON-formatted event logs using Linux command-line tools.

Figure 1: Practical Detection and Response Lab Exercise

```

11/23/2022-12:38:34.624866  [**] [1:12345:3] GET on wire [**] [Classification: (
null)] [Priority: 3] [TCP] 172.21.224.2:49652 -> 142.250.1.139:80
11/23/2022-12:38:58.958203  [**] [1:12345:3] GET on wire [**] [Classification: (
null)] [Priority: 3] [TCP] 172.21.224.2:58494 -> 142.250.1.102:80
analyst@6b9e43cb2c8d:~$ ls -l /var/log/suricata
total 16
-rw-r--r-- 1 root root 1418 May 20 12:36 eve.json
-rw-r--r-- 1 root root 292 May 20 12:36 fast.log
-rw-r--r-- 1 root root 3241 May 20 12:36 stats.log
-rw-r--r-- 1 root root 1512 May 20 12:36 suricata.log
analyst@6b9e43cb2c8d:~$ cat /var/log/suricata/eve.json
{"timestamp":"2022-11-23T12:38:34.624866+0000","flow_id":2172860802693269,"pcap
cnt":70,"event_type":"alert","src_ip":"172.21.224.2","src_port":49652,"dest_ip":
"142.250.1.139","dest_port":80,"proto":"TCP","tx_id":0,"alert":{"action":"allowe
d","gid":1,"signature_id":12345,"rev":3,"signature":"GET on wire","category":"","
severity":3},"http":{"hostname":"opensource.google.com","url":"/","http_user_ag
ent":"curl/7.74.0","http_content_type":"text/html","http_method":"GET","protocol
":"HTTP/1.1","status":301,"redirect":"https://opensource.google/","length":223},
"app_proto":"http","flow":{"pkts_toserver":4,"pkts_toclient":3,"bytes_toserver":
357,"bytes_toclient":788,"start":"2022-11-23T12:38:34.620693+0000"}}
{"timestamp":"2022-11-23T12:38:58.958203+0000","flow_id":388155863897332,"pcap_c
nt":151,"event_type":"alert","src_ip":"172.21.224.2","src_port":58494,"dest_ip":
"142.250.1.102","dest_port":80,"proto":"TCP","tx_id":0,"alert":{"action":"allowe
d","gid":1,"signature_id":12345,"rev":3,"signature":"GET on wire","category":"","
severity":3},"http":{"hostname":"opensource.google.com","url":"/","http_user_ag
ent":"curl/7.74.0","http_content_type":"text/html","http_method":"GET","protocol
":"HTTP/1.1","status":301,"redirect":"https://opensource.google/","length":223},
"app_proto":"http","flow":{"pkts_toserver":4,"pkts_toclient":3,"bytes_toserver":
357,"bytes_toclient":797,"start":"2022-11-23T12:38:58.955636+0000"}}
analyst@6b9e43cb2c8d:~$ jq . /var/log/suricata/eve.json

```

The `eve.json` file is the standard and main Suricata log file and contains a lot more data than the `fast.log` file. This data is stored in a JSON format, which makes it much more useful for analysis and processing by other applications.

1. Use the `cat` command to display the entries in the `eve.json` file:

```
cat /var/log/suricata/eve.json
```

The output returns the raw content of the file. You'll notice that there is a lot of data returned that is not easy to understand in this format.

2. Use the `jq` command to display the entries in an improved format:

```
jq . /var/log/suricata/eve.json | less
```

Note: You can use the lowercase `f` and `b` keys to move forward or backward through the output. Also, if you enter a command incorrectly and it fails to return to the command-line prompt, you can press `CTRL+C` to stop the process and force the shell to return to the command-line prompt.

Figure 2: Practical Detection and Response Lab Exercise

```

analyst@6b9e43cb2c8d:~$ cat custom.rules
alert http $HOME_NET any -> $EXTERNAL_NET any (msg:"GET on wire"; flow:establish
ed,to_server; content:"GET"; http_method; sid:12345; rev:3;)
analyst@6b9e43cb2c8d:~$ ls -l /var/log/suricata
ls: cannot access '/var/log/suricata': No such file or directory
analyst@6b9e43cb2c8d:~$ ls -l /var/log/suricata
total 0
analyst@6b9e43cb2c8d:~$ sudo suricata -r sample.pcap -S custom.rules -k none
20/5/2026 -- 12:36:23 - <Notice> - This is Suricata version 6.0.1 RELEASE runnin
g in USER mode
20/5/2026 -- 12:36:24 - <Notice> - all 2 packet processing threads, 4 management
threads initialized, engine started.
20/5/2026 -- 12:36:24 - <Notice> - Signal Received. Stopping engine.
20/5/2026 -- 12:36:24 - <Notice> - Pcap-file module read 1 files, 200 packets, 5
4238 bytes
analyst@6b9e43cb2c8d:~$ ls -l /var/log/suricata
total 16
-rw-r--r-- 1 root root 1418 May 20 12:36 eve.json
-rw-r--r-- 1 root root 292 May 20 12:36 fast.log
-rw-r--r-- 1 root root 3241 May 20 12:36 stats.log
-rw-r--r-- 1 root root 1512 May 20 12:36 suricata.log
analyst@6b9e43cb2c8d:~$ cat /var/log/suricata/fast.log
11/23/2022-12:38:34.624866  [**] [1:12345:3] GET on wire [**] [Classification: (
null)] [Priority: 3] [TCP] 172.21.224.2:49652 -> 142.250.1.139:80
11/23/2022-12:38:58.958203  [**] [1:12345:3] GET on wire [**] [Classification: (
null)] [Priority: 3] [TCP] 172.21.224.2:58494 -> 142.250.1.102:80
analyst@6b9e43cb2c8d:~$ ls -l /var/log/suricata
total 16
-rw-r--r-- 1 root root 1418 May 20 12:36 eve.json
-rw-r--r-- 1 root root 292 May 20 12:36 fast.log

```

The `eve.json` file is the standard and main Suricata log file and contains a lot more data than the `fast.log` file. This data is stored in a JSON format, which makes it much more useful for analysis and processing by other applications.

1. Use the `cat` command to display the entries in the `eve.json` file:

```
cat /var/log/suricata/eve.json
```

The output returns the raw content of the file. You'll notice that there is a lot of data returned that is not easy to understand in this format.

2. Use the `jq` command to display the entries in an improved format:

```
jq . /var/log/suricata/eve.json | less
```

Note: You can use the lowercase `f` and `b` keys to move forward or backward through the output. Also, if you enter a command incorrectly and it fails to return to the command-line prompt, you can press `CTRL+C` to stop the process and force the shell to return to the command-line prompt.

Figure 3: Practical Detection and Response Lab Exercise

```
    "redirect": "https://opensource.google/",
    "length": 223
  },
  "app_proto": "http",
  "flow": {
    "pkts_toserver": 4,
    "pkts_toclient": 3,
    "bytes_toserver": 357,
    "bytes_toclient": 797,
    "start": "2022-11-23T12:38:58.955636+0000"
  }
}
analyst@6b9e43cb2c8d:~$ jq . /var/log/suricata/eve.json | less
{
  "timestamp": "2022-11-23T12:38:34.624866+0000",
  "flow_id": 2172860802693269,
  "pcap_cnt": 70,
  "event_type": "alert",
  "src_ip": "172.21.224.2",
  "src_port": 49652,
  "dest_ip": "142.250.1.139",
  "dest_port": 80,
  "proto": "TCP",
  "tx_id": 0,
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 12345,
    "rev": 3,
    "signature": "GET on wire",
```

10. Conclusion

The Detection and Response – Sound the Alarm course provided valuable theoretical knowledge and practical cybersecurity experience. The hands-on labs strengthened my ability to analyze network traffic, investigate alerts, interpret security logs, and understand how cybersecurity analysts respond to threats in operational environments.

The practical exposure to intrusion detection systems, Linux command-line analysis, and incident response workflows significantly improved my technical confidence and problem-solving abilities in cybersecurity operations.

Overall, the course prepared me with foundational knowledge and practical skills required for cybersecurity monitoring, threat detection, and incident response roles within modern security operations environments.